RISE Student Research Grant Proposal

Part A: Project Proposal Summary

A.1 Proposed title

A Supervised Principal Component Analysis Based Clustering Approach to Find Correlation between Alzheimer's Disease and Different Clinical and Lifestyle Variables

A.2 Research area

Health Informatics (Alzheimer's Disease), Bioinformatics (Single-cell and Spatial Transcriptomics (ST) Data Analysis), Artificial Intelligence, Machine Learning

A.3 Research summary (max. 150 words)

Alzheimer's disease (AD), a slowly progressing neurodegenerative disorder, is the most common form of dementia[1]. Single-cell atlases have proven to be very informative for unraveling the etiology of diseases like COVID-19[2]. However, for AD, single-cell studies have not fully captured the full spectrum of cognitive decline and end-stage cognition.

Mathys *et al.*[3] generated a single-cell transcriptomics atlas of the aged human brain with varying degrees of AD pathology and cognitive impairment to understand the biology behind it. However, the correlation between AD and different lifestyle and clinical variables has been established based on pre-dividing the cell types, which failed to find any significant association with non-AD-pathology variables.

In this research, we propose to use Supervised Principal Component Analysis (SPCA), a popular dimensionality-reduction technique for supervised machine learning, to cluster the cell types only after they show significant gene expression changes with both AD-pathology and non-AD-pathology variables. This approach will provide better insights into the more important genes associated with AD and provide a better visualization for future analysis.

A.4 Project duration (in months)	10
A.5 Total project cost (in BDT)	1,00,000

Part B: Research Project Details

B.1 Background and statement of the problem (Word limit: Max. 200)

[1] Wilson, R.S., Segawa, E., Boyle, P.A., Anagnos, S.E., Hizel, L.P., and Bennett, D.A. (2012). The Natural History of Cognitive Decline in Alzheimer's Disease. Psychol. Aging 27, 1008–1017. https://doi.org/10.1037/A0029857

[2] Delorey, T.M., Ziegler, C.G.K., Heimberg, G., Normand, R., Yang, Y., Segerstolpe, A˚., Abbondanza, D., Fleming, S.J., Subramanian, A., Montoro, D.T., et al. (2021). COVID-19 tissue atlases reveal SARS-CoV-2 pathology and cellular targets. Nat 595, 107–113. https://doi.org/10.1038/s41586-021-03570-8

[3] Mathys et al., 2023, Cell 186, 4365–4385 September 28, 2023, ª 2023 The Authors. Published by Elsevier Inc. https://doi.org/10.1016/j.cell.2023.08.039

Alzheimer's disease (AD) starts with mild memory loss and culminates in severe impairment of broad executive and cognitive functions[4]. The pathological hallmarks of AD encompass a range of clinical presentations and cellular pathologies. As a result, analyses of the aged human brain across larger unbiased cohorts are needed to capture this variability.

Mathys *et al.*[3] covered more than 2.3 million cells in their single-cell transcriptomics atlas, working on post-mortem tissue samples from 427 individuals having varying degrees of AD pathology- from no cognitive impairment to mild cognitive impairment and AD dementia. Bair *et al.*[5] described a technique called supervised principal component analysis that can be applied to regression problems where the number of predictors greatly exceeds the number of observations and identify which predictor variables are the most important based on their association with the outcome.

To determine how individual cell types are affected by different clinical and lifestyle variables, it is necessary to analyze their differential gene expression with regard to each variable, and then cluster the similar genes together for further analysis. In our research, we propose to analyze the single-cell atlas data using supervised principal component analysis to find the association between genes and different lifestyle and clinical variables more accurately.

B.2 Specific objectives (Word limit: Max. 200)

The main objectives of this research are as follows:

1. To analyze the differential gene expression for at least 36 clinical and lifestyle variables using supervised principal component analysis.
2. To cluster and visualize the most important genes associated with both AD-pathology and non-AD-pathology variables.
3. To create a pre-processed dataset which can be used for further analysis such as the temporal changes in gene expression across multiple stages of AD progression.

B.3 Expected outcomes (Word limit: Max. 150)

The expected outcomes of this research project are as follows:

1. A robust supervised principal component analysis model based on gene expression data of 427 individuals.
2. Insights into which cell types are particularly vulnerable to AD pathology burden, and how their loss can explain cognitive impairment.
3. Understanding the biology behind association of AD-pathology and non-AD-pathology variables with clinical and lifestyle variables.

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[4] Scheltens, P., Blennow, K., Breteler, M.M.B., De Strooper, B., Frisoni, G.B., Salloway, S., and Van Der Flier, W.M. (2016). Alzheimer's disease. Lancet.. https://doi.org/10.1016/S0140-6736(15)01124-1

[5] Eric Bair, Trevor Hastie, Debashis Paul & Robert Tibshirani (2006) Prediction by Supervised Principal Components, Journal of the American Statistical Association, 101:473, 119-137, DOI: 10.1198/016214505000000628

B.4 Methodology (World limit: Max. 500)
Data Collection:

We will be using the following four related datasets for our analysis:

1. The MIT ROSMAP Single-Nucleus Multiomics Study (MIT_ROSMAP_Multiomics)[6]
2. The Religious Orders Study and Memory and Aging Project (ROSMAP) Study[7]
3. Longitudinal, epidemiologic clinical-pathologic cohort study of aging and Alzheimer's disease (AD)[8]
4. Longitudinal, epidemiologic clinical-pathologic cohort study of common chronic conditions of aging with emphasis on decline in cognitive and motor function and risk of AD[8]

We have already obtained access to the first two studies mentioned above and our access requests are currently under process for the third and the fourth.

Data Cleaning and Feature Extraction:

The mentioned four studies contain gene expression data of more than 2.3 million nuclei from 427 individuals in total. We will use a randomization algorithm for sampling 10% of the provided data and analyze their association with the top 3000 genes ranked on their variability.

Model Creation:

Since supervised principal component analysis is an extremely effective technique for regression problems where the number of predictors (genes) greatly exceeds the number of observations (cells), we will apply this model to extract the key principal components and their loading based on their association with the outcome.

Training and Validation:

We will divide our dataset into training set, validation set and independent test sets. The training set will be used to learn the classification model while validation set will be used for necessary hyperparameter tuning. We will then assess the quality of our model using the independent test set. We will compute the Z-score, P value, and Cross-Validation Error for our model and compare them with the existing models in the literature.

Challenges:

1. Our dataset containing more than 2.3 million nuclei is very large. To mitigate this, we plan to sample only 10% of the data due to our computational limitations, which may not reflect the

[6] MIT_ROSMAP_Multiomics. https://doi.org/10.7303/syn2580853
[7] The Religious Orders Study and Memory and Aging Project (ROSMAP) Study. https://doi.org/10.7303/syn2580853
[8] Bennett DA, Buchman AS, Boyle PA, Barnes LL, Wilson RS, Schneider JA. Religious Orders Study and Rush Memory and Aging Project. J Alzheimers Dis. 2018;64(s1):S161-S189. doi: 10.3233/JAD-179939. PMID: 29865057; PMCID: PMC6380522.

full performance of our model properly.

2. Training a machine learning model requires a high-performance CPU, GPU, significant amount of RAM and fast storage (SSD hard drive). As we have a large volume of data (e.g. 131 GB for the first two datasets), transferring the data to some online computation platform (e.g. Google Colab) may not be efficient. Therefore, we plan to use an onsite high-performance computer for this research.

Part C: Proposed Grant Amount Summary

(*Detail budget must be submitted in a separate file.*)

SN	Budget description	Total amount (BDT)
(i)	Resources	1,00,000
(ii)	Miscellaneous	0
(iii)	Conference Registration and Participation Expenses (maximum BDT 10,000)	0
	Grand total amount (i+ii+iii)	**1,00,000**
(in words) One lac only.		

Part D: Student Details

D.1 Applicant-1	
Name	Md. Ishrak Ahsan
Department	Computer Science and Engineering
Email address	ishrak26@gmail.com
Contact no.	01826071969
Research experience (Word limit: Max. 100)	1. Developed a desktop application imitating Club Football Transfer Market using JavaFX from June 2021 to July 2021.[9] 2. Co-developed a multiplayer hardware game using ATMega32 and Arduino Uno from January 2023 to February 2023.[10] 3. Working on Single-cell Transcriptomics Data Analysis, in collaboration with Dr. Md. Abul Hassan Samee, Baylor College of Medicine, USA from November 2023 to date.

D.2 Applicant-2	
Name	
Department	
Email address	
Contact No.	
Research experience (Word limit: Max. 100)	
Add more applicants if necessary.	

[9] https://youtu.be/KRCxGvgQvro?si=31mws1EHv-YgkZzA
[10] https://youtu.be/NY4GrnJCqsM?si=ZkKjrz7ZkLyEwd0W